

From Release Kinetics to Field Performance: A Critical Review of Slow- and Controlled-Release Fertilizers, Nutrient-Use Efficiency, Environmental Benefits, and Scale-Up Challenges

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Abstract

Slow- and controlled-release fertilizers (SRFs and CRFs) are increasingly used to improve nutrient-use efficiency and reduce nutrient losses, yet slower release does not inherently translate into superior field performance. This critical review synthesizes 130 included publications published from 2020 to 2026, supplemented by three foundational sources used only for historical and technology-readiness context, using a performance-first framework that links material architecture, material release, root-zone nutrient availability, plant uptake, yield, nutrient loss, environmental impact, cost, and industrial readiness. The evidence shows that dissolution, diffusion, osmotic transport, swelling, coating degradation, ion exchange, mineral reactions, and biological processes can dominate under different conditions. Nitrogen systems have the strongest evidence base, phosphorus is moderately supported, potassium and mineral systems remain more heterogeneous, and evidence for micronutrients is comparatively limited. A crop–supply synchronization concept and a five-level field-specific technology-readiness framework for slow- and controlled-release fertilizers (TRL-SRF) are proposed as conceptual tools requiring independent validation; TRL-SRF is not equivalent to conventional nine-level TRL systems.

Environmental gains must also be weighed against coating persistence, microplastics, embodied inputs, and cost. The optimum fertilizer is therefore the one whose plant-available nutrient-delivery profile best matches crop demand under realistic soil and climatic conditions, rather than the formulation with the longest release duration.

Keywords: Slow-release fertilizer; Controlled-release fertilizer; Nutrient-use efficiency; Nutrient-release kinetics; Sustainable agriculture; Enhanced-efficiency fertilizers

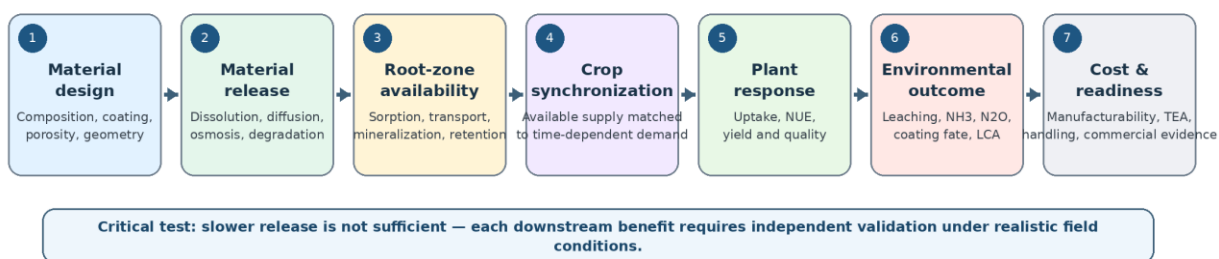
Highlights

- Release rate alone is insufficient to predict agronomic performance.
- Crop–release synchronization is the key criterion for fertilizer design.
- Soil conditions can substantially alter laboratory-derived release profiles.
- Scale-up, cost, and coating sustainability remain major commercialization barriers.

Graphical abstract

Performance-first framework for slow- and controlled-release fertilizers

Design is valuable only when release control survives soil processes and improves agronomic, environmental, and industrial outcomes.



1. Introduction

1.1. Problem definition and performance gap

Fertilizer efficiency is constrained by a structural mismatch between the rapid availability of many conventional soluble fertilizers and the time-dependent nutrient demand of crops. Nitrogen can be lost through ammonia volatilization, nitrate leaching, and nitrous oxide emissions; phosphorus can become immobilized by sorption or precipitation; and potassium can be displaced below the effective rooting zone or supplied in forms whose dissolution is too slow for peak crop demand. Enhanced-efficiency fertilizers attempt to reduce this mismatch by controlling the rate, timing, or biochemical transformation of nutrient supply. Recent reviews and meta-analyses show substantial agronomic promise, but they also reveal strong dependence on crop, soil, climate, formulation, and management context (Jariwala et al., 2022; Jiang et al., 2022; Liu et al., 2023; Pan et al., 2024; Priya, Sarkar, et al., 2024; Siudak et al., 2026; Vejan et al., 2021; Yang et al., 2021; Zhang, Wang, et al., 2026; Zhang et al., 2024; Zhu et al., 2020).

Slow-release fertilizers (SRFs) and controlled-release fertilizers (CRFs) are often grouped together, yet their governing principles are not identical. Mineral-derived SRFs may rely on inherently slow dissolution, ion exchange, or reaction-limited nutrient liberation, whereas CRFs more commonly use a coating or engineered matrix to regulate water ingress and nutrient diffusion. The distinction matters because the same apparent release duration can arise from different rate-controlling mechanisms, with different sensitivities to temperature, moisture, ionic composition, abrasion, and microbial activity (Duan et al., 2023; Lakshani et al., 2023; Lawrencina et al., 2021; Liu et al., 2022; Priya, Sarkar, et al., 2024; Vejan et al., 2021).

1.2. Why slower release is not automatically better

The central limitation of much of the literature is therefore conceptual rather than merely technological: a slower dissolution curve measured in water does not, by itself, demonstrate higher nutrient-use efficiency (NUE), higher yield, lower environmental burden, or economic superiority in the field. Studies that explicitly connect fertilizer formulation with crop response, nutrient losses, and long-term soil function are more informative than release tests alone. Long-duration field evidence also shows that one environmental benefit can coexist with another burden; for example, reducing ammonia loss does not necessarily guarantee a reduction in nitrous oxide emissions (Gao et al., 2024; Guo et al., 2026; Guo et al., 2024; Liu, Chen, et al., 2025; Lü et al., 2023; Ren et al., 2026; Wang et al., 2024; Zhang et al., 2025).

Accordingly, this review is organized around a performance-first question: under which combinations of fertilizer chemistry, release architecture, soil conditions, crop demand, environmental exposure, and manufacturing constraints does slower nutrient release translate into genuinely improved field performance? The review evaluates the evidence chain from material design and release mechanism to soil interaction, nutrient availability, plant uptake, yield, nutrient loss, environmental impact, cost, and industrial readiness. This framing intentionally shifts the discussion from cataloging materials toward identifying the conditions under which release control is agronomically and commercially meaningful.

2. Review Methodology and Evidence Framework

2.1. Literature-search strategy and evidence base

This critical integrative review used Google Scholar as the primary discovery platform. The final literature update was conducted on 13 September 2026, with the analytic window restricted

to publications from 2020 through 2026. The search program comprised 38 topic-specific Boolean families (B001–B038), with up to 20 candidate references retained per Boolean family before cross-query consolidation. Representative expressions included: (i) ("slow-release fertilizer" OR "controlled-release fertilizer" OR "enhanced-efficiency fertilizer") AND ("nutrient release" OR "release kinetics" OR "nutrient use efficiency" OR NUE OR "crop yield" OR "agronomic performance" OR "environmental impact" OR sustainability OR scale-up OR commercialization); (ii) ("controlled-release fertilizer" OR "coated fertilizer") AND (diffusion OR diffusivity OR "diffusion-controlled") AND ("nutrient release" OR permeability); (iii) ("slow-release fertilizer" OR "controlled-release fertilizer") AND (dissolution OR "dissolution-controlled release") AND (nutrient OR fertilizer OR release); and (iv) nutrient-specific combinations for nitrogen/urea, phosphorus/struvite/hydroxyapatite, potassium/mineral sources, agronomic performance, environmental losses, coating fate, manufacturing, scale-up, and commercialization. No formal language filter was imposed at the query stage, although the final corpus is predominantly English-language because of database coverage and the requirement for sufficiently detailed, traceable reporting.

Corpus consolidation was auditable through an archived bibliographic matrix. The structured matrix records 266 Boolean–reference associations and 121 DOI/title-deduplicated unique records with section mapping and evidence notes. Final targeted updating and reference checking expanded the manuscript corpus to 130 included publications from 2020–2026. Three older foundational sources were added separately to anchor controlled-release theory, enhanced-efficiency fertilizer terminology, and conventional technology-readiness context; these sources were not counted in the 130-study analytic corpus.

A publication was included when it contributed directly to at least one predefined evidence domain: fertilizer class or terminology; material architecture; release mechanism or kinetic

interpretation; soil/root-zone behavior; crop response, nutrient uptake, yield, or NUE; nutrient-loss or environmental outcomes; manufacturing/scale-up; techno-economic or life-cycle assessment; or industrial/commercial validation. Records were excluded when they were outside the date window for the analytic corpus, duplicated by DOI/title, peripheral to fertilizer-release performance, lacked traceable publication metadata, or provided no interpretable evidence for any of the predefined domains. When several studies addressed the same question, emphasis was placed on methodological rigor, mechanistic characterization, field validation, multi-season duration, direct environmental measurement, industrial-scale evidence, and meta-analytic coverage. Accordingly, the term "included publications" is used throughout instead of "priority publications" to avoid implying an unstructured subjective ranking.

Because this review was conceived as a critical integrative review rather than a prospectively registered systematic review, raw-hit counts before deduplication and the exact numbers excluded at each screening step were not logged contemporaneously. Those counts are therefore not retrospectively invented. The available auditable counts and the distinction between the structured search matrix and the final included corpus are reported explicitly below, and the search-audit workbook is supplied as Supplementary File S1.

The auditable corpus-construction flow is summarized in Figure S1.

Auditable construction of the 2020-2026 review corpus

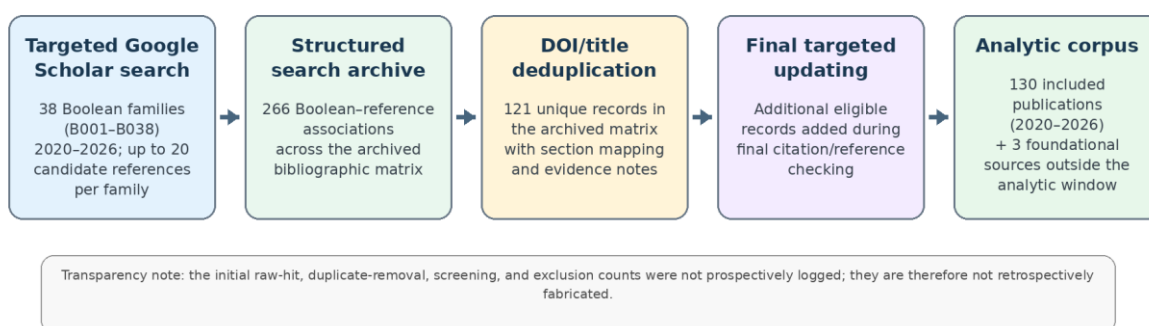


Figure S1. Auditable construction of the review corpus from targeted Boolean searches to the final 130-publication analytic set.

Note. The archived bibliographic matrix contains 38 Boolean families, 266 Boolean-reference associations, and 121 unique DOI/title-deduplicated records. The final manuscript corpus contains 130 included publications after targeted updating. Raw initial-hit and exclusion counts were not prospectively logged and are not reconstructed retrospectively.

2.2. Evidence hierarchy

Evidence was classified by experimental level: (i) aqueous or solution release tests, (ii) soil incubation, (iii) pot or greenhouse studies, (iv) single-season field trials, (v) multi-season field trials, (vi) pilot-scale manufacturing or application, and (vii) industrial or commercial validation. This hierarchy is not a quality ranking in the conventional sense; rather, it reflects increasing exposure to interacting variables that can alter release and performance. A formulation with excellent reproducibility in water may fail in soil, while a field-effective material may show release behavior that cannot be explained by a single idealized kinetic model (Bonetti et al., 2022; Duan et al., 2023; Lipin et al., 2023; Nguyen et al., 2021; Ransom et al., 2020; Swify et al., 2023).

2.3. Critical assessment framework

The critical assessment framework used throughout the review is shown in Figure 1. It links material architecture to material release, soil and rhizosphere interactions, root-zone nutrient availability, plant uptake, yield and NUE, nutrient loss, environmental impact, and cost/readiness. The framework represents a hypothesized causal evidence chain in which each downstream relationship requires independent empirical validation; the review does not assume that causality has been demonstrated across every transition. For example, an improved diffusion barrier can demonstrate material-release control, but it does not by itself establish root-zone availability or yield response; likewise, a yield response does not establish lower life-cycle impact. Throughout the manuscript, three rates are therefore kept conceptually distinct: (i) material release rate from the fertilizer, (ii) soil/root-zone plant-available nutrient supply rate after transport, sorption, precipitation, mineralization, immobilization, and other transformations, and (iii) plant uptake rate.

Figure 1 integrates the evidence chain used throughout this review and makes explicit that each downstream claim requires its own validation step.

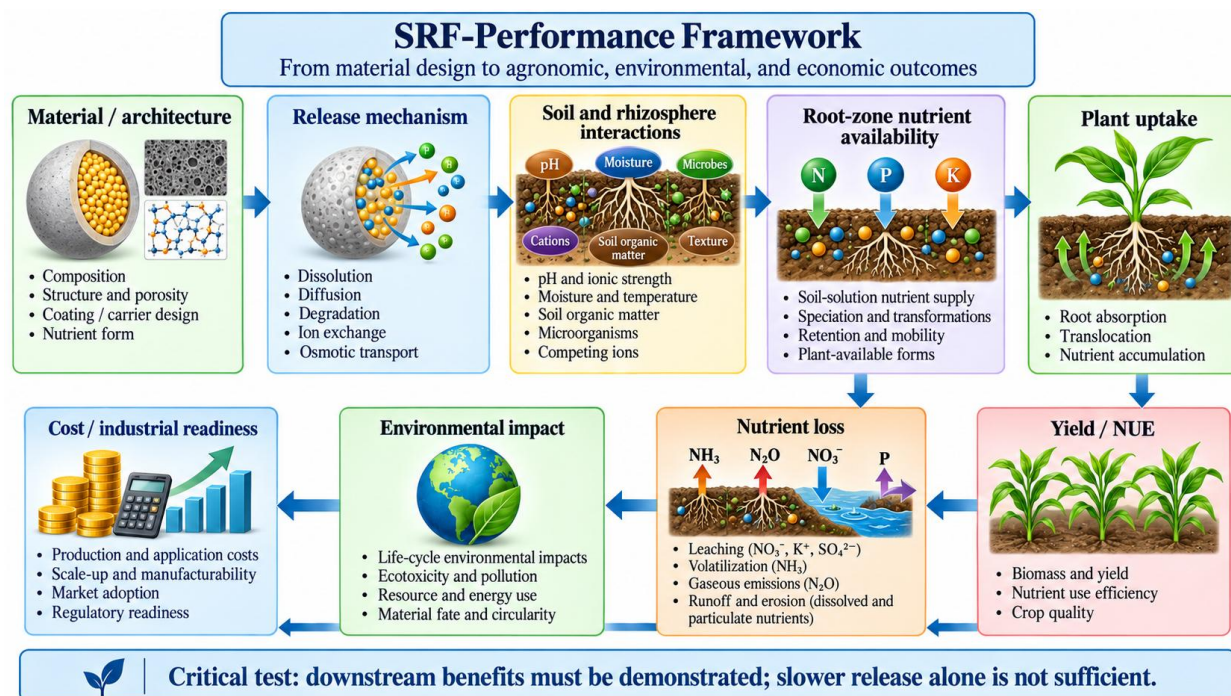


Figure 1. Conceptual framework linking fertilizer architecture to agronomic, environmental, and industrial performance.

Note. Conceptual framework synthesized by the author from release-mechanism, material-design, agronomic, and environmental frameworks discussed by Duan et al. (2023), Jariwala et al. (2022), Lawrencina et al. (2021), and Siudak et al. (2026).

2.4. Mechanistic versus empirical evidence

The review further distinguishes mechanistic evidence from empirical curve fitting. Release models are evaluated according to whether their assumptions are physically plausible for the tested material, whether parameters remain transferable across temperatures and soil conditions, and whether model validation is independent of the data used for fitting (Lakshani et al., 2023; Lipin et al., 2023; Mendonca Cidreira et al., 2026; Ren et al., 2024; Shi et al., 2022). This prevents the

common error of using a high coefficient of determination as proof of a specific transport mechanism.

3. Terminology, Classification, and Functional Boundaries

3.1. Functional definitions

Terminology is a major source of inconsistency in this field. Conventional soluble fertilizers are designed to dissolve rapidly and provide immediate nutrient availability. SRFs reduce nutrient availability through intrinsic material properties or matrix effects, whereas CRFs use an intentionally engineered barrier, membrane, or encapsulating phase to regulate release. Stabilized fertilizers constitute a separate functional class because they primarily slow biochemical transformations, such as urea hydrolysis or nitrification, rather than physically controlling nutrient release. Hybrid products can combine these functions and should be described by the dominant mechanisms rather than by marketing terminology alone (Lawrencina et al., 2021; Liu et al., 2022; Priya, Sarkar, et al., 2024; Siudak et al., 2026; Vejan et al., 2021). These functional distinctions are consistent with foundational controlled-release and enhanced-efficiency fertilizer frameworks, while the present review applies them to the current 2020–2026 evidence base (Shaviv, 2001; Trenkel, 2010).

3.2. Mechanism-based classification

Table 1 summarizes the functional boundaries among the major enhanced-efficiency fertilizer classes. This distinction matters in practice because release duration, nutrient chemistry, and field response cannot be interpreted without identifying what controls mass transfer. Mineral matrices, zeolites, layered double hydroxides (LDHs), struvite, biochar composites, hydrogels, and polymer-coated granules may all be described as 'slow release', but their controlling processes can range from dissolution and ion exchange to

diffusion, swelling, and coating degradation (Buates & Imai, 2021; Hertzberger et al., 2020; Legese et al., 2024; Li et al., 2024; Liu et al., 2024; Maghsoodi, Najafi, et al., 2020; Marcińczyk & Oleszczuk, 2022; Mihok et al., 2020; Roy et al., 2023).

Table 1. Definitions and functional classification of conventional, slow-release, controlled-release, stabilized, and hybrid fertilizers.

Class	Primary control	Typical examples	Critical limitation
Conventional soluble	Rapid dissolution	Urea, soluble NPK, KCl	High early nutrient availability can exceed crop demand.
Slow-release (SRF)	Intrinsic dissolution, ion exchange, or matrix transport	Struvite, phosphate minerals, zeolites, K-bearing silicates	Release may be too slow or strongly soil-dependent.
Controlled-release (CRF)	Engineered barrier or encapsulation	Polymer-coated urea, polymer-sulfur products	Coating defects, cost, and persistence.
Stabilized	Inhibition of biochemical transformation	Urease- or nitrification-inhibited fertilizers	Does not necessarily control physical nutrient release.
Hybrid / multifunctional	Combined barrier, sorption, water retention, or stabilization	Hydrogels, biochar composites, coated organominerals	Mechanisms and nutrient-specific release can be difficult to disentangle.

Note. Classification is mechanism-based. A commercial label should not substitute for identification of the controlling release process.

Figure 2 complements Table 1 by showing the hierarchical relationship among enhanced-efficiency fertilizer classes and the material families most commonly associated with each mechanism.

Functional classification of enhanced-efficiency fertilizers

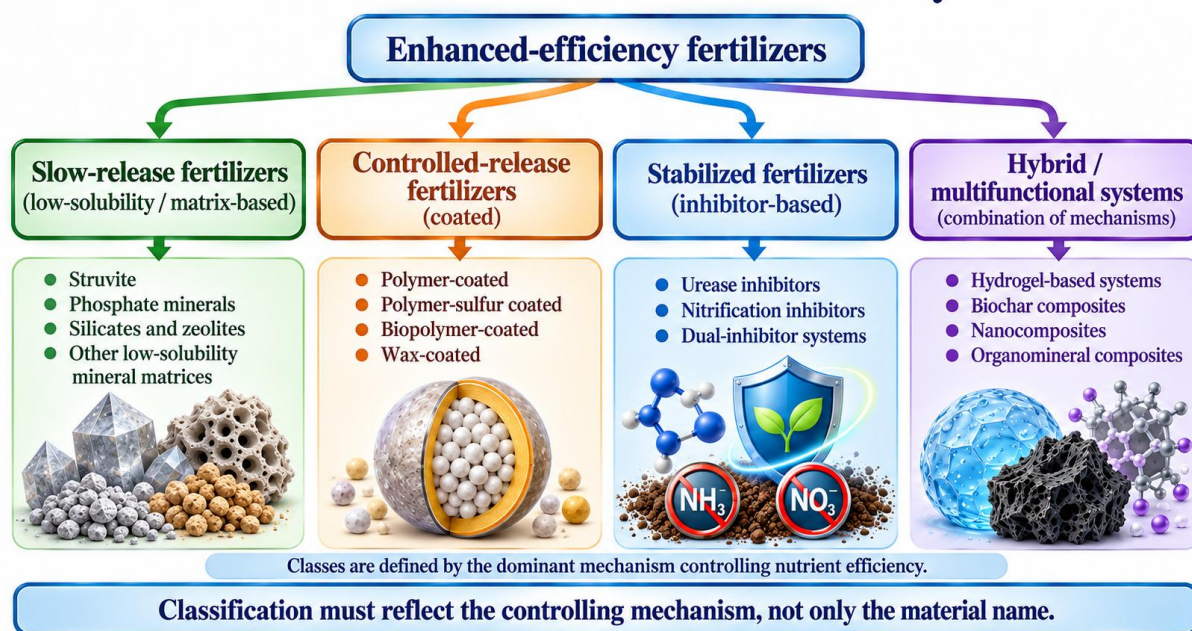


Figure 2. Functional classification of enhanced-efficiency fertilizer technologies.

Note. Classification adapted by the author from the functional distinctions and material categories discussed by Lawrencina et al. (2021), Priya, Sarkar, et al. (2024), Siudak et al. (2026), and Vejan et al. (2021).

3.3. Coated, matrix-based, and hybrid systems

A second boundary concerns coated versus matrix-based systems. In coated systems, a relatively soluble nutrient core is separated from the environment by a layer whose thickness, permeability, defects, and degradation determine transport. In matrix systems, nutrient species are dispersed or chemically incorporated within a solid phase, and release can depend on coupled dissolution, pore transport, ion exchange, or matrix degradation. Composite systems blur this boundary, particularly when biochar, clays, hydrogels, or biodegradable polymers simultaneously retain water, sorb nutrients, and provide diffusion resistance (Bauli et al., 2021; Bi et al., 2020;

Bora et al., 2023; Bortoletto-Santos et al., 2020; Han et al., 2025; Hu et al., 2024; Ji et al., 2024; Li & Zhang, 2023; Priya, Jha, et al., 2024; Sarhan et al., 2024; Shen et al., 2021; Song et al., 2024; Xu et al., 2026).

4. Fundamentals of Nutrient Release

4.1. Governing release mechanisms

Nutrient release emerges from transport, reaction, and structural change. Dissolution-controlled systems are governed by the solubility and reactive surface of the nutrient-bearing phase; diffusion-controlled systems depend on concentration gradients and effective diffusivity through coatings or matrices; osmotic systems depend on water ingress and internal pressure; swelling-controlled systems change permeability as hydrophilic networks expand; and degradable coatings evolve structurally with time. Ion exchange, adsorption/desorption, mineral transformation, and biological activity can add further rate limitations (Bi et al., 2020; Bortoletto-Santos et al., 2020; Duan et al., 2023; Lakshani et al., 2023; Lawrencina et al., 2021; Li et al., 2024; Lipin et al., 2023; Maghsoodi, Najafi, et al., 2020; Mendonca Cidreira et al., 2026; Mihok et al., 2020; Priya, Sarkar, et al., 2024; Sakhno et al., 2022; Zhao et al., 2020).

Figure 3 summarizes the principal physical, chemical, and biological routes that can control the observed release profile.

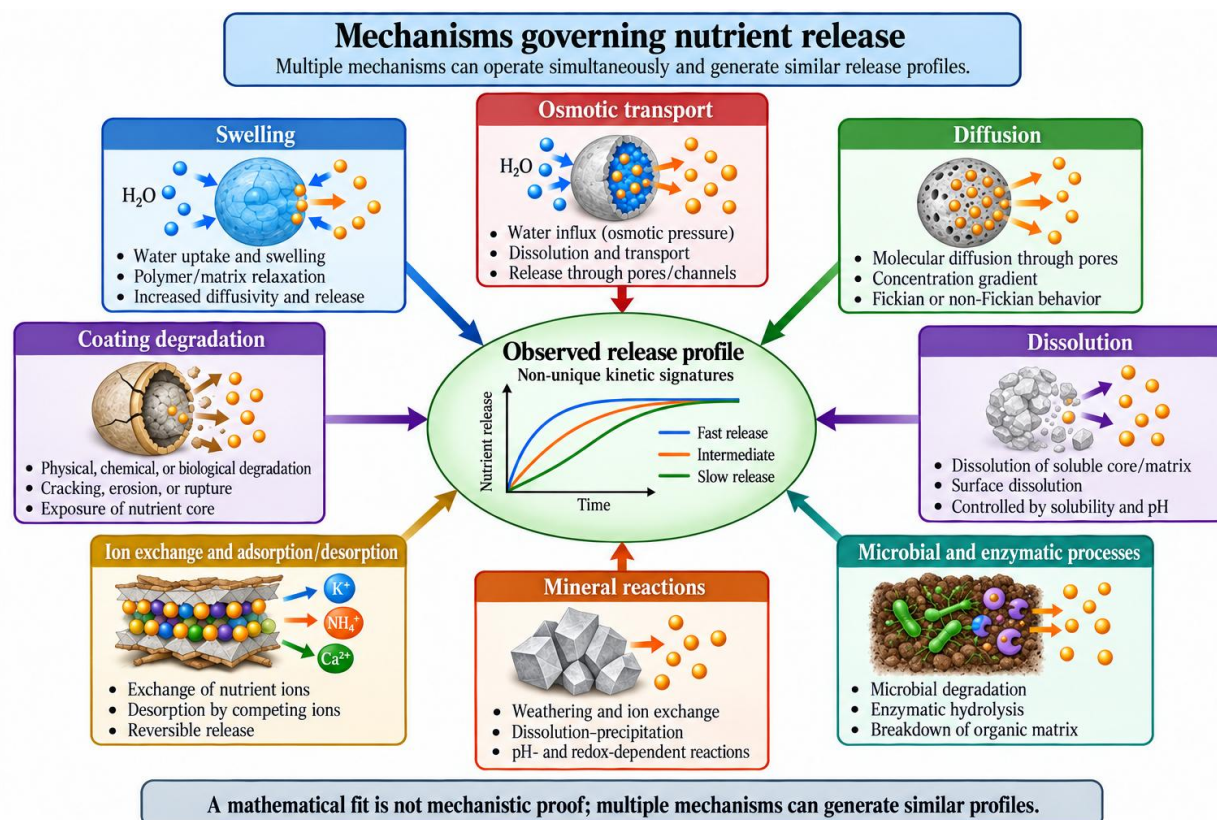


Figure 3. Mechanisms of nutrient release: dissolution, diffusion, osmotic transport, swelling, degradation, ion exchange, mineral reaction, and biological processes.

Note. Mechanism map adapted by the author from Duan et al. (2023), Lakshani et al. (2023), Lawrencina et al. (2021), and Lipin et al. (2023).

4.2. Quantitative release descriptors

A useful first descriptor is the fractional release, $F(t) = M_t/M_\infty$, where M_t is the cumulative nutrient mass released at time t , and M_∞ is the potentially releasable mass. Common models include zero-order, first-order, Higuchi-type diffusion, Korsmeyer-Peppas relations, sigmoidal functions, and mechanistic diffusion-reaction models. Table 2 summarizes these forms. Their main value is comparative when test conditions are controlled; do not treat parameter values as intrinsic

material constants unless you explicitly incorporate temperature, water activity, geometry, and chemical boundary conditions (Duan et al., 2023; Lakshani et al., 2023; Lipin et al., 2023).

$$F(t) = M_t / M_{\infty} \quad (1)$$

$$F(t) = k_0 t \quad (2)$$

$$F(t) = 1 - \exp(-k_1 t) \quad (3)$$

$$F(t) = kH t^{1/2} \quad (4)$$

$$F(t) = kKP t^n \quad (5)$$

Table 2. Mathematical models commonly used to describe nutrient-release kinetics and their interpretive limits.

Model	Representative form	Main use	Critical caution
Zero-order	$F(t)=k_0 t$	Idealized constant release	Often valid only over a limited interval.
First-order	$F(t)=1-\exp(-k_1 t)$	Concentration-dependent release	Parameter changes with boundary conditions.
Higuchi	$F(t)=kH\sqrt{t}$	Diffusion-dominated matrix release	Assumes simplified geometry and diffusivity.
Korsmeyer–Peppas	$F(t)=k t^n$	Empirical transport classification	Exponent n is not proof of a unique mechanism.
Sigmoidal	$F(t)=1/[1+\exp(-k(t-t_{50}))]$	Lag + accelerated + terminal release	Descriptive unless linked to physical transitions.
Mechanistic diffusion–reaction	Mass balance + transport/reaction terms	Scale and condition prediction	Requires independently validated parameters.

Note. $F(t)$ denotes fractional nutrient release. Select models based on physical plausibility and independent validation, not goodness of fit alone.

4.3. Linking models to physical evidence

The strongest mechanistic studies couple release measurements with structural or chemical evidence. Examples include work on coating uniformity, polyurethane nanocomposites, molecular-scale zeolite behavior, biochar-based release, and mineral transformation, where release is interpreted together with material properties rather than by curve shape alone (Ali et al., 2023;

Azeem, 2025; Azeem, 2026; Bortoletto-Santos et al., 2020; El Messbahi et al., 2026; Mendonca Cidreira et al., 2026; Mihok et al., 2020). Figure 3 therefore emphasizes that multiple mechanisms may contribute to the same observed profile.

4.4. Boundary-condition dependence

A critical consequence is that 'controlled release' should be treated as a boundary-condition-dependent property. A coating that behaves as a diffusion barrier in water can become defect-controlled after drying, mechanically damaged during handling, or biologically altered in soil. Similarly, a low-solubility phosphate or potassium mineral can shift from reaction-limited to transport-limited behavior as weathering products accumulate. Mechanistic claims should therefore be tested under more than one chemically relevant environment whenever the intended application is field agriculture.

5. Materials and Fertilizer Architectures

5.1. Polymer and biopolymer coatings

Material design determines both the fertilizer release mechanism and its environmental burden. Synthetic polymers provide tunable permeability and mechanical integrity, but persistence and microplastic generation remain concerns. Biobased polyurethanes, lignin blends, starch, cellulose, chitosan, alginate, collagen-derived networks, and biomass waxes are being developed to retain release control while improving feedstock renewability and end-of-life behavior (Arafa et al., 2022; Channab et al., 2023; dos Santos et al., 2021; Fertahi et al., 2021; Ji et al., 2024; Kontárová et al., 2022; Kosanovich et al., 2026; Meng et al., 2023; Priya, Jha, et al., 2024; Purba et al., 2026; Ravindran et al., 2025; Shi et al., 2026; Song et al., 2024; Su et al., 2024; Uzoh et al., 2026; Xu et al., 2026; Yao et al., 2025). Importantly, biobased and biodegradable are not

interchangeable descriptors: biobased identifies feedstock origin, whereas biodegradable requires verified breakdown under specified environmental conditions and timescales. A biobased coating can persist, and a biodegradable material need not be predominantly biobased.

5.2. Hydrogels and water-retaining matrices

Hydrogels and superabsorbent matrices can provide dual functionality by storing water and delaying nutrient transport. However, their performance depends on swelling stability, ionic sensitivity, mechanical strength, and how well water-retention benefits persist under repeated wetting-drying cycles. The literature increasingly combines biopolymers with nanocellulose, clays, or other reinforcements to improve structure while maintaining biodegradability (Bauli et al., 2021; Bora et al., 2023; Hu et al., 2024; Li & Zhang, 2023; Perera et al., 2026; Priya, Jha, et al., 2024; Purba et al., 2026; Sarhan et al., 2024; Shen et al., 2021; Song et al., 2024; Xu et al., 2026).

5.3. Mineral and carbonaceous architectures

Mineral and carbonaceous architectures offer a different design space. Zeolites, LDHs, biochar, hydrochar, struvite, hydroxyapatite, phosphate minerals, and K-bearing silicates can act as nutrient reservoirs, sorbents, or reactive phases rather than inert barriers. Their release behavior is therefore inseparable from mineralogy, ion exchange, dissolution equilibria, and soil chemistry (Basak et al., 2023; Bonetti et al., 2022; Buates & Imai, 2021; Cheng et al., 2023; Everaert et al., 2022; Fachini et al., 2022; Gonçalves & Roque, 2026; Han et al., 2025; Hertzberger et al., 2020; Inácio et al., 2025; Kaur et al., 2026; Kokulan et al., 2024; Legese et al., 2024; Li et al., 2024; Li, Xu, et al., 2023; Liu et al., 2024; Maghsoodi, Ghodszad, et al., 2020; Maghsoodi, Najafi, et al., 2020; Marcińczyk & Oleszczuk, 2022; Pereira, 2026; Roy et al., 2023; Sakhno et al., 2022; Subramani & Balakrishnan, 2026; Taskin & Gunes, 2023; Teles et al., 2020; Wang et al., 2022).

Table 3 compares the principal material families according to dominant release control, advantages, and limitations. The critical design implication is that release duration alone cannot identify the 'best' material. Consider cost, coating mass, raw-material availability, mechanical robustness, biodegradation, manufacturing yield, and the ability to tune release to crop demand simultaneously. Materials that require complex synthesis, high solvent consumption, or difficult recovery steps may be scientifically interesting but remain far from commercial relevance.

Table 3. Main material families used in slow- and controlled-release fertilizers, dominant release controls, strengths, and constraints.

Material family	Dominant release control	Strengths	Main constraints
Synthetic polymers	Diffusion through coating	Highly tunable; established coating routes	Persistence, microplastics, petrochemical inputs
Biobased polyurethane/wax	Diffusion + degradation	Renewable feedstocks; hot-melt potential	Feedstock variability; degradation validation
Starch/cellulose/lignin systems	Swelling, diffusion, degradation	Biodegradable feedstocks	Moisture sensitivity; mechanical strength
Hydrogels	Swelling-controlled diffusion	Water retention + nutrient delivery	Ionic sensitivity; repeated-cycle stability
Biochar/hydrochar composites	Sorption/desorption + pore diffusion	Circular feedstocks; soil-conditioning potential	Heterogeneous chemistry; variable release
Zeolites/aluminosilicates	Ion exchange + diffusion	High cation retention; mineral stability	Nutrient loading and exchange selectivity
LDHs/clays	Intercalation/exchange + dissolution	Strong anion/cation hosting	Soil pH and competing ions
Struvite/hydroxyapatite	Mineral dissolution	Recovered P potential; low loss risk	Dissolution can be crop/soil limited
K-bearing silicates	Weathering/dissolution	Alternative K resources	Often slow kinetics; processing may be energy intensive
Organomineral/composite	Multiple coupled mechanisms	Multi-nutrient and circular design	Mechanistic complexity and standardization

Note. Dominant controls may shift with soil chemistry, coating degradation, particle damage, and water regime.

6. Nutrient-Specific Slow-Release Systems

6.1. Nitrogen

Nitrogen dominates the CRF literature because urea is highly soluble, widely used, and vulnerable to multiple loss pathways. Polymer-coated, sulfur-modified, hydrogel-based, biochar-based, and blended controlled-release urea formulations have been evaluated across rice, maize, cotton, wheat, sunflower, oilseed rape, vegetables, and other systems (Chen et al., 2020; Etminani-Esfahani et al., 2025; Gao et al., 2024; Geng et al., 2026; Guo et al., 2024; Ji et al., 2022; Jiang et al., 2022; Kontárová et al., 2022; Liu, Zhang, et al., 2025; Liu et al., 2023; Lü et al., 2023; Ma et al., 2026; Ma et al., 2024; Moon et al., 2026; Qiao et al., 2025; Ren et al., 2024; Ren et al., 2025; Ren et al., 2026; Shi et al., 2022; Tang et al., 2021; Tian et al., 2021; Xiao et al., 2026; Yang et al., 2021; Zhang et al., 2022; Zhang et al., 2025; Zhang et al., 2024; Zheng et al., 2020; Zheng et al., 2026; Zhu et al., 2020). The key agronomic question is not whether release can be delayed, but whether the delay reduces early-season surplus without creating late-season nutrient limitation.

6.2. Phosphorus

Phosphorus requires a different logic because the dominant constraint is often low mobility and chemical fixation rather than excessive solubility alone. Struvite, LDHs, hydroxyapatite, modified rock phosphate, and nano-enabled phosphates aim to moderate solution P activity while maintaining plant availability. Field and review evidence suggests that slow P delivery can reduce loss risk and sustain crop uptake, but performance is especially sensitive to soil pH, sorption capacity, rhizosphere chemistry, and the ability of roots or microbes to mobilize the source (Buates & Imai, 2021; Everaert et al., 2022; Gonçalves & Roque, 2026; Hertzberger et al., 2020; Inácio et al., 2025; Kaur et al., 2026; Kokulan et al., 2024; Li et al., 2024; Liu et al., 2024; Maghsoodi,

Ghodsad, et al., 2020; Perera et al., 2026; Roy et al., 2023; Sakhno et al., 2022; Subramani & Balakrishnan, 2026; Taskin & Gunes, 2023; Teles et al., 2020; Wang et al., 2023).

6.3. Potassium

Potassium SRFs include controlled-release KCl and K₂SO₄, zeolites, biochar-derived K systems, and processed silicate minerals such as feldspars, micas, syenites, and glauconitic materials. These alternatives span a very wide reactivity range. Coated soluble salts primarily regulate mass transfer, whereas mineral-derived K fertilizers depend on structural breakdown and dissolution; therefore, the same term 'slow release' covers fundamentally different design problems (Ait Harrouch et al., 2026; Basak et al., 2023; Chen et al., 2024; Chen et al., 2026; Cheng et al., 2023; Crusciol et al., 2020; El Messbahi et al., 2026; Fachini et al., 2022; Li et al., 2020; Liu, Chen, et al., 2025; Peixoto et al., 2022; Pereira & Fonseca, 2025; Pereira, da Cunha Fonseca, et al., 2026; Pereira, Santos, et al., 2025; Qu et al., 2024; Qu et al., 2025).

6.4. Secondary nutrients, micronutrients, and NPK

Secondary nutrients and micronutrients are less extensively represented, but multifunctional formulations are expanding. Mg-enriched biochar-based compound fertilizers and Zn-exchanged clays illustrate how nutrient delivery can be integrated with sorptive or soil-conditioning functions (Cyriac et al., 2024; Li et al., 2026). Multi-nutrient organomineral and NPK hydrogel systems further complicate evaluation because one nutrient's release profile can differ substantially from another's within the same granule (Bora et al., 2023; Crusciol et al., 2020; Ngo et al., 2022). Table 4 organizes slow-release strategies by nutrient and identifies the dominant mechanistic constraints for each group. The evidence density is therefore intentionally graded in this review: conclusions

are strongest for N, moderately supported for P, emerging and more heterogeneous for K/mineral systems, and insufficient for broad generalization to micronutrients.

Table 4. *Slow-release systems according to nutrient, representative technologies, dominant constraints, and performance criteria.*

Nutrient	Representative systems	Dominant constraint	Performance criterion
N	Coated urea, hydrogel urea, and blended controlled-release urea (CRU)	Volatilization, leaching, N ₂ O, timing	Synchrony with crop N demand and loss reduction
P	Struvite, LDH-P, hydroxyapatite, modified rock phosphate	Fixation, low mobility, dissolution	Sustained plant-available P without excess runoff risk
K	Coated KCl/K ₂ SO ₄ , zeolites, biochar, feldspar/mica/syenite	Leaching vs mineral dissolution	K supply matched to crop uptake and soil buffering
S/Ca/Mg	Compound granules, mineral matrices	Nutrient-specific solubility	Balanced multi-nutrient availability
Micronutrients	Zn-exchanged clays, nanocomposites	Narrow sufficiency-toxicity range	Controlled availability without accumulation
NPK/multi-nutrient	Hydrogels, organominerals, composites	Different release rates within one product	Independent tracking of each nutrient flux

Note. The appropriate release criterion is nutrient-specific; identical release durations do not imply equivalent agronomic value.

7. Factors Governing Release under Real Soil Conditions

7.1. Temperature, water, and transport

Release tests performed in water are useful for screening, but the field environment introduces coupled thermal, hydraulic, chemical, and biological controls. Temperature alters diffusivity, polymer mobility, reaction rates, and microbial activity; water content controls both solute transport and the continuity of aqueous pathways; and ionic strength changes swelling, ion exchange, and activity gradients. Placement and particle size also modify the local boundary layer and root access. Ransom et al. (2020) demonstrated the importance of placement and temperature,

while soil-texture studies and field comparisons show that the same fertilizer can behave differently across soil environments (Swify et al., 2023; Zhang, Wang, et al., 2026).

7.2. pH, mineralogy, and reactive soils

Soil pH and mineralogy are particularly important for P- and K-bearing SRFs. Acid-base conditions control phosphate speciation, apatite dissolution, struvite stability, and silicate-mineral weathering; clays and oxides can either retain released nutrients or reduce their effective diffusion toward roots. Comparisons between hydroponic and soil systems demonstrate why release behavior cannot be extrapolated directly from simplified media (El Messbahi et al., 2026; Gonçalves & Roque, 2026; Maghsoodi, Ghodszad, et al., 2020; Peixoto et al., 2022; Perera et al., 2026; Roy et al., 2023; Sakhno et al., 2022; Teles et al., 2020).

7.3. Biological and management controls

Biological and management variables are equally important. Controlled-release fertilization can alter microbial co-occurrence patterns and N-cycling functions, while straw return, irrigation regime, mulching, and deep placement modify both nutrient transport and crop access (Chen et al., 2026; Gao et al., 2024; Ji et al., 2022; Li et al., 2022; Ma et al., 2024; Qu et al., 2024; Qu et al., 2025; Tang et al., 2021; Zheng et al., 2026). These interactions justify the multivariable representation in Figure 4: field release depends on material properties and the environment, not a fixed duration printed on a product label.

Table 5 consolidates the main soil and environmental variables to consider when transferring release data from simplified tests to field conditions.

Table 5. Effects of soil, environmental, and product variables on nutrient release and laboratory-to-field transferability.

Variable	Expected influence	Why it matters for transferability
Temperature	Changes diffusivity, polymer mobility, reaction and microbial rates	Constant-temperature tests may misrepresent seasonal release.
Soil moisture	Controls water ingress and aqueous transport continuity	Drying can interrupt diffusion or damage coatings.
pH	Changes mineral solubility, polymer ionization, and nutrient speciation	Especially important for P minerals and responsive hydrogels.
Ionic strength	Changes swelling, exchange, and activity gradients	Saline soils may suppress or accelerate release.
Texture/mineralogy	Controls sorption, tortuosity, and retention	Released nutrient may not remain plant-available.
Organic matter/microbes	Affects degradation and nutrient transformation	Biodegradable materials depend on biological context.
Placement	Changes root contact, moisture, and diffusion path	Deep banding can rival or exceed formulation effects.
Particle/coating geometry	Controls surface area, thickness, and defect probability	Scale-up must preserve relevant geometry.

Note. Variables act interactively; single-factor tests are screening tools rather than complete simulations of field behavior.

Figure 4 visualizes these interacting controls around the field release profile and highlights why release duration is not an invariant material property.

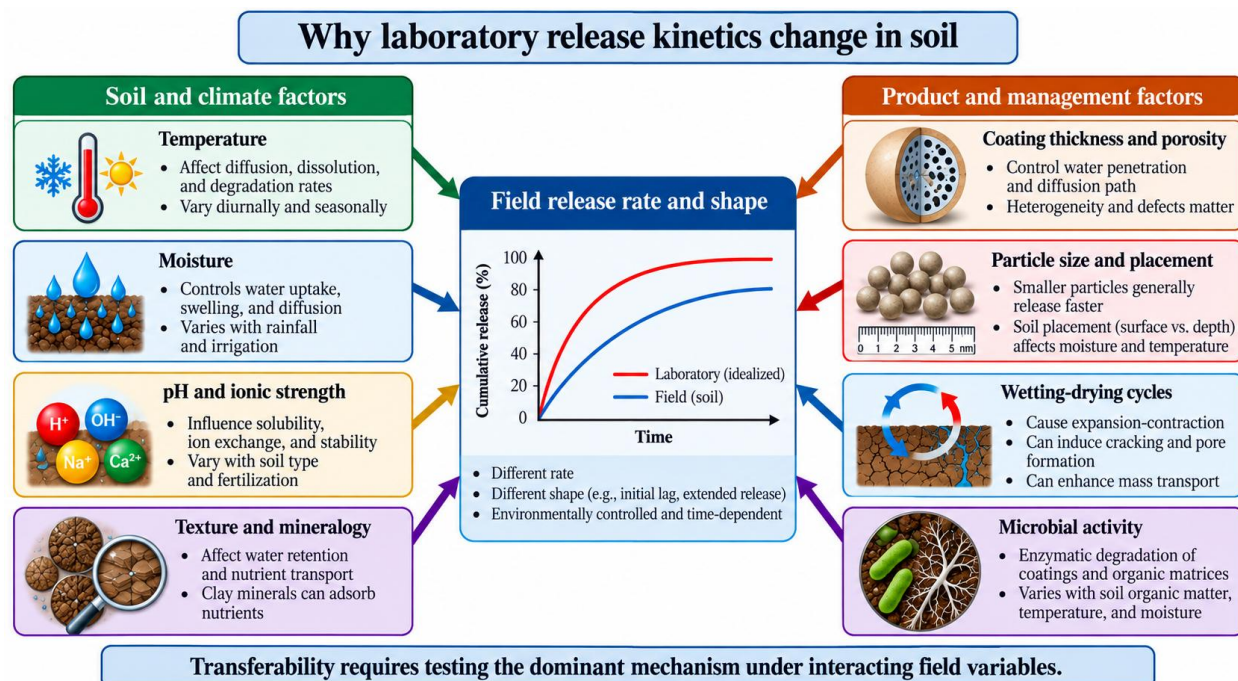


Figure 4. Effect of coating properties, soil conditions, and management on nutrient-release kinetics.

Note. Conceptual synthesis adapted by the author from Ransom et al. (2020), Swify et al. (2023), and Zhang, Wang, et al. (2026).

7.4. Validation envelopes

The design implication is that laboratory protocols should reproduce the dominant field stressors expected for the target use. For irrigated systems, repeated wetting and temperature cycles may be more informative than a single constant-temperature water test. For arid soils, low water activity and salt accumulation may dominate. For tropical acid soils, mineral dissolution and sorption processes may be more important than coating permeability. A robust SRF/CRF therefore requires an application-specific validation envelope rather than a universal release specification.

8. Matching Nutrient Release with Crop Demand

8.1. Crop-demand dynamics

The most important performance concept in this review is synchronization. Crop demand is dynamic: early vegetative demand can be low, uptake may accelerate sharply during biomass accumulation, and demand later declines or shifts among nutrients. A fertilizer that releases very slowly can therefore be as poorly synchronized as one that releases too rapidly. Recent work explicitly linking controlled-release N supply to plant-soil synchrony, nutrient remobilization, and oilseed-rape uptake supports moving beyond simple release-duration targets (Ma et al., 2026; Moon et al., 2026; Ren et al., 2024; Xiao et al., 2026).

8.2. Synchronization metric

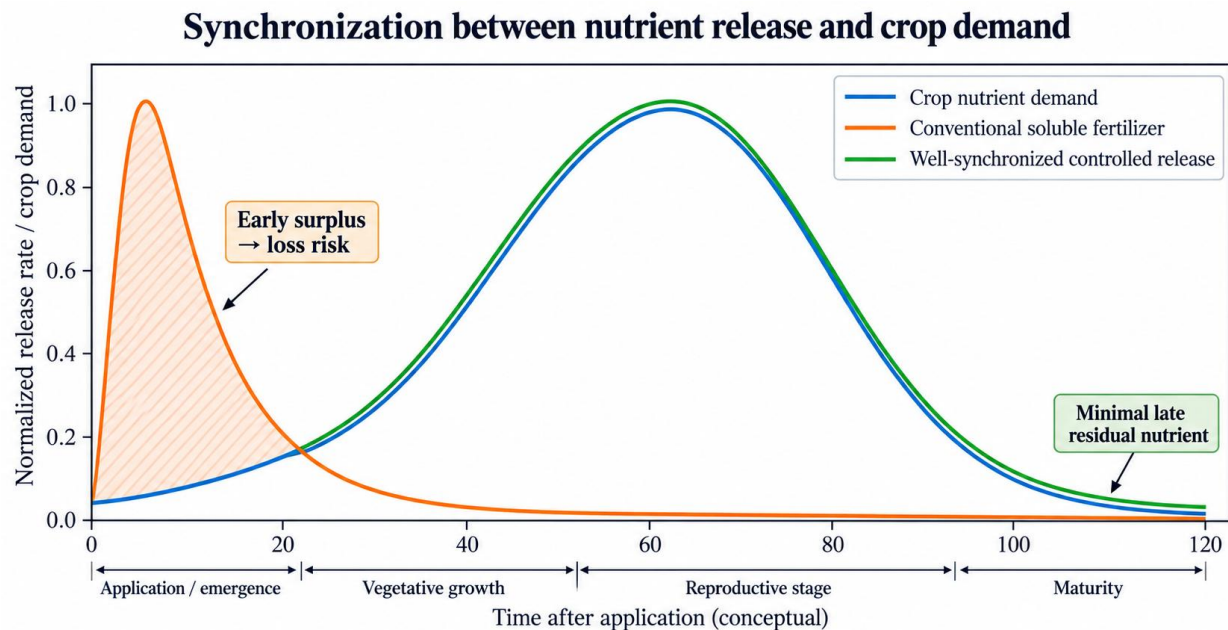
Figure 5 illustrates the distinction between rapid conventional release, crop demand, and a well-synchronized controlled-release profile. For field interpretation, however, material release $R_m(t)$ should not be equated automatically with plant-available supply. Let $A(t)$ denote the non-negative root-zone available nutrient-supply rate over the interval $t_0 \leq t \leq t_f$, where $A(t)$ includes fertilizer-derived nutrient that remains plant-available after soil transformations plus endogenous soil-derived supply from mineralization, residual fertility, native pools, irrigation water, and biologically mediated transformations. Let $D(t)$ denote the non-negative crop nutrient-demand or uptake-requirement rate over the same interval. To compare temporal shapes independently of total dose, define normalized functions $\hat{A}(t) = A(t)/\int A(t)dt$ and $\hat{D}(t) = D(t)/\int D(t)dt$, so each function has unit area and units of inverse time. The proposed synchronization index is then $S = 1 - \{\int |\hat{A}(t) - \hat{D}(t)|dt / \int [\hat{A}(t) + \hat{D}(t)]dt\}$. Because both normalized functions integrate to one, the denominator equals 2 and $0 \leq S \leq 1$. $S = 1$ indicates identical temporal supply and

demand shapes; $S = 0$ represents complete temporal separation of the two normalized profiles. Unavailable or negative nutrient pools are not entered into $A(t)$; carryover and residual soil fertility enter only to the extent that they contribute to non-negative plant-available supply within $[t_0, t_f]$. If root-zone availability is not measured, normalized material release $R_m(t)$ may be used only as an explicitly labeled proxy. The index is a proposed conceptual comparator, not a validated replacement for measured NUE, yield, nutrient recovery, or nutrient-loss endpoints, and it requires independent external validation.

$$S = 1 - \left\{ \int |\hat{A}(t) - \hat{D}(t)| dt / \int [\hat{A}(t) + \hat{D}(t)] dt \right\} \quad (6)$$

Worked hypothetical example. If the integrated absolute difference between two unit-area normalized profiles over the crop-growth interval is 0.30, the denominator is 2.00 and $S = 1 - (0.30/2.00) = 0.85$. This value indicates strong temporal similarity, but it does not demonstrate adequate nutrient quantity, high NUE, or superior yield; those outcomes must be measured independently.

Figure 5 shows the intended simplified interpretation of Equation (6): both early nutrient surpluses and excessively delayed supply reduce synchronization. In the schematic figure, fertilizer release is used as a visual proxy for root-zone available supply; field application of the index should use $A(t)$ whenever soil-supply data are available.



Agronomic performance improves when nutrient release better matches crop demand over time.

Figure 5. Synchronization between nutrient-release profile and crop nutrient-demand curve.

Note. Conceptual synchronization curves adapted by the author from the plant–soil synchrony concepts discussed by Ma et al. (2026) and Xiao et al. (2026). Curves are schematic and not fitted to a specific dataset.

8.3. Blended and placement strategies

Synchronization also explains why blended strategies often perform well. Combining conventional and controlled-release urea can provide an immediately available fraction while sustaining later supply, and meta-analyses indicate that the optimum ratio depends on crop and management context rather than a universal formulation (Liu, Zhang, et al., 2025; Tang et al., 2021; Zhang et al., 2022; Zheng et al., 2020). Similarly, placement can be as important as release control; global evidence indicates that deep fertilization can sometimes outperform enhanced-

efficiency fertilizer alone, emphasizing that technology and agronomy should be evaluated together (Wu et al., 2025).

8.4. Crop-specific design

Crop-specific fertilizer design should therefore use phenology, rooting depth, expected soil mineralization, irrigation, and climate to define the target nutrient-delivery curve. Precision agriculture could extend this logic by spatially varying rate or formulation, but the value of such complexity must be tested against simpler alternatives such as split application, deep placement, or blended products.

9. Agronomic Performance and Nutrient-Use Efficiency

9.1. Meta-analytic evidence

Agronomic performance is the decisive test of release-control technology. Meta-analyses across rice, cotton, maize, and broader crop datasets generally report positive average effects of controlled-release N on yield and NUE, but they also identify substantial heterogeneity. Soil, climate, crop, coating type, application rate, and the proportion of CRF in blended programs act as moderators, which is why average treatment effects should not be interpreted as guaranteed field benefits (Jiang et al., 2022; Liu, Zhang, et al., 2025; Liu et al., 2023; Pan et al., 2024; Qiao et al., 2025; Tian et al., 2021; Yang et al., 2021; Zhang et al., 2022; Zhang, Wang, et al., 2026; Zhang et al., 2024; Zhu et al., 2020). These meta-analytic conclusions are strongest for N-based systems; the smaller and more heterogeneous P and K evidence bases should not be assumed to have the same effect size, mechanism maturity, or transferability, and micronutrient systems remain insufficient for broad cross-nutrient inference.

9.2. Long-term field evidence

Long-term and field-scale studies are particularly informative. Controlled-release K has been associated with changes in K availability, nutrient uptake, and maize ecosystem properties over multiple years; controlled-release N studies report effects on yield stability, soil quality, microbial functions, and greenhouse-gas behavior; and field studies with struvite show that lower-solubility P sources can maintain crop production while changing loss risk (Guo et al., 2026; Li et al., 2020; Liu, Chen, et al., 2025; Ma et al., 2024; Wang et al., 2023; Wang et al., 2024). These findings reinforce that fertilizer value accumulates through both current-season crop response and residual soil effects.

9.3. Yield, uptake, and NUE are not equivalent

The relationships among release rate, NUE, and yield are not equivalent. A slower release can increase NUE without increasing yield if the crop is already near its yield ceiling; it can increase yield without proportionally improving recovery efficiency if soil nutrient supply is high; and it can fail on both metrics when release is delayed beyond periods of active uptake. Field studies in rice, peanut, sunflower, soybean, wheat, and other crops demonstrate the need to report multiple efficiency metrics rather than yield alone (Chen et al., 2020; Geng et al., 2026; Qu et al., 2025; Ren et al., 2025; Ren et al., 2026; Zhang et al., 2025; Zheng et al., 2026).

Figure 6 places agronomic evidence within a laboratory-to-commercial transition and identifies the main sources of performance loss at each step.

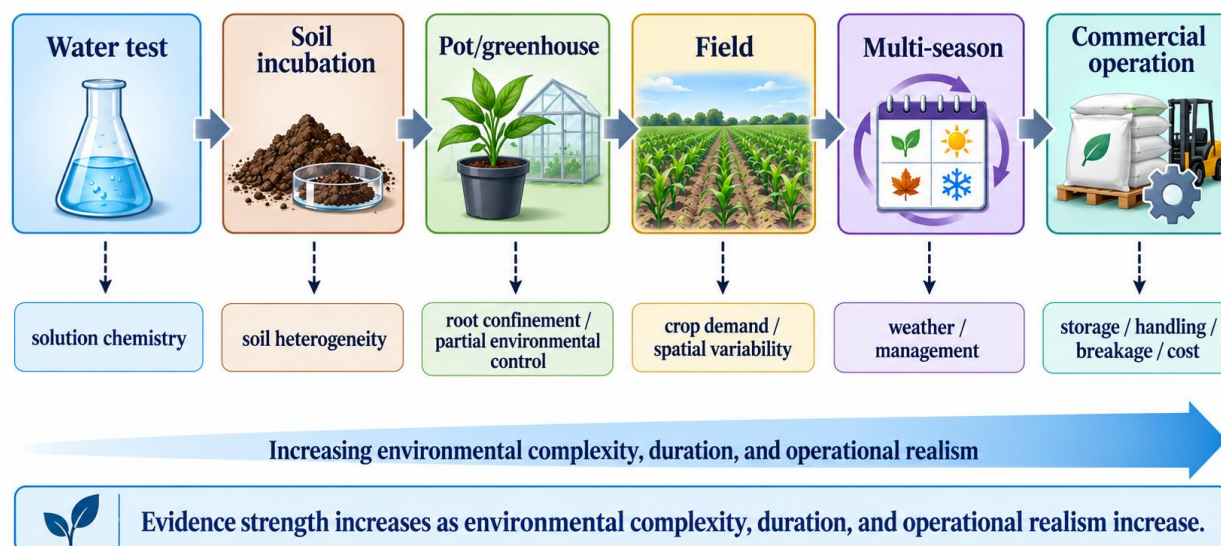


Figure 6. Laboratory-to-field scale transition and factors causing performance loss.

Note. Evidence-transition framework adapted by the author from pilot, scale-up, field, and long-term studies synthesized in this review, including Bonetti et al. (2022) and Nguyen et al. (2021).

9.4. Representative field comparisons

Table 6 presents representative field evidence using directional outcomes rather than pooled numerical effect sizes. This avoids false precision across studies with different soils, crops, durations, and application rates. The table also highlights an evidence-quality principle: a study that reports yield, nutrient uptake, loss pathways, and multiple seasons provides more decision-relevant information than a single-season yield comparison.

Table 6. Representative field studies comparing controlled/slow-release and conventional fertilizer strategies.

Study / crop	Technology	Main field signal	Critical interpretation
Chen et al. (2020) / rice	Controlled-release urea	Yield, NUE, and soil-fertility response	Benefits must be interpreted with local N rate and soil conditions.
Li et al. (2020) / maize	Controlled-release KCl	Long-term K availability and nutrient absorption	Shows value of multi-season nutrient balance.

Study / crop	Technology	Main field signal	Critical interpretation
Gao et al. (2024) / long-term cropping	Blended controlled-release urea + straw	Yield, greenhouse-gas (GHG) emissions, and economic outcomes	Integrates agronomic and environmental endpoints.
Guo et al. (2024) / summer maize	Controlled-release urea	Lower NH ₃ , improved yield, higher N ₂ O risk	Clear example of pathway trade-off.
Kokulan et al. (2024) / field corn	Struvite	Maintained yield with lower P-loss risk	Demonstrates P-specific environmental value.
Qu et al. (2025) / soybean	Deep-banded controlled-release KCl	Higher productivity and K uptake	Placement and formulation interact.
Ren et al. (2025) / sunflower	Controlled-release fertilizer	Higher NUE/yield via N allocation and photosynthesis	Links physiological response to field performance.
Wang et al. (2023) / rice–wheat	Struvite	Yield, uptake, and residual soil P over two years	Multi-season evidence improves confidence.
Zhang et al. (2025) / wheat	Two-time controlled-release fertilization	Yield, NUE, and N ₂ O response	Timing remains relevant even with CRF.
Zheng et al. (2026) / rice	Controlled-release N + straw return	Yield, soil nutrients, and NUE response	System management modifies fertilizer effect.

Note. Directional summaries are used because rates, soils, climates, and experimental designs are not directly comparable across studies.

10. Environmental Performance and Sustainability Trade-Offs

10.1. Nutrient-loss mitigation

Environmental performance is often cited as a major advantage of SRFs and CRFs because lower early nutrient concentrations can reduce leaching, volatilization, and runoff. Meta-analysis and field studies support reductions in reactive-N losses under many conditions, but the direction and magnitude of response vary with soil, climate, formulation, and management (Jiang et al., 2022; Lü et al., 2023; Pan et al., 2024; Ren et al., 2026; Shi et al., 2022; Tang et al., 2021; Yang et al., 2021; Zhang et al., 2025; Zhang et al., 2024). The environmental case is therefore strongest when losses are measured directly rather than inferred from slower release.

10.2. Nitrogen pathway trade-offs

Trade-offs are particularly important for nitrogen. Long-term controlled-release urea can reduce ammonia volatilization while shifting the conditions that regulate N₂O production, meaning that a technology can improve one atmospheric loss pathway and worsen another (Guo et al., 2024; Lü et al., 2023; Wang et al., 2024). Likewise, lower application frequency may reduce field operations while the coating itself adds embodied energy or persistent material. Figure 7 summarizes these competing dimensions.

10.3. Polymer persistence and microplastics

Polymer persistence and microplastics have become central concerns. Recent studies identify mechanisms of microplastic formation from coated fertilizers, assess risks to soil-crop systems, evaluate photodegradable resin residues, and examine biodegradation and long-term aging of biobased and petroleum-based polyurethanes (Bhattacharjee et al., 2025; Jo et al., 2026; Li et al., 2025; Tian et al., 2026; Tian et al., 2024). These studies demonstrate why 'biobased' and 'biodegradable' should not be treated as interchangeable claims: degradation rate, products, soil exposure, and functional lifetime must all be characterized.

Figure 7 makes these trade-offs explicit and places life-cycle and field performance at the center of the environmental comparison.

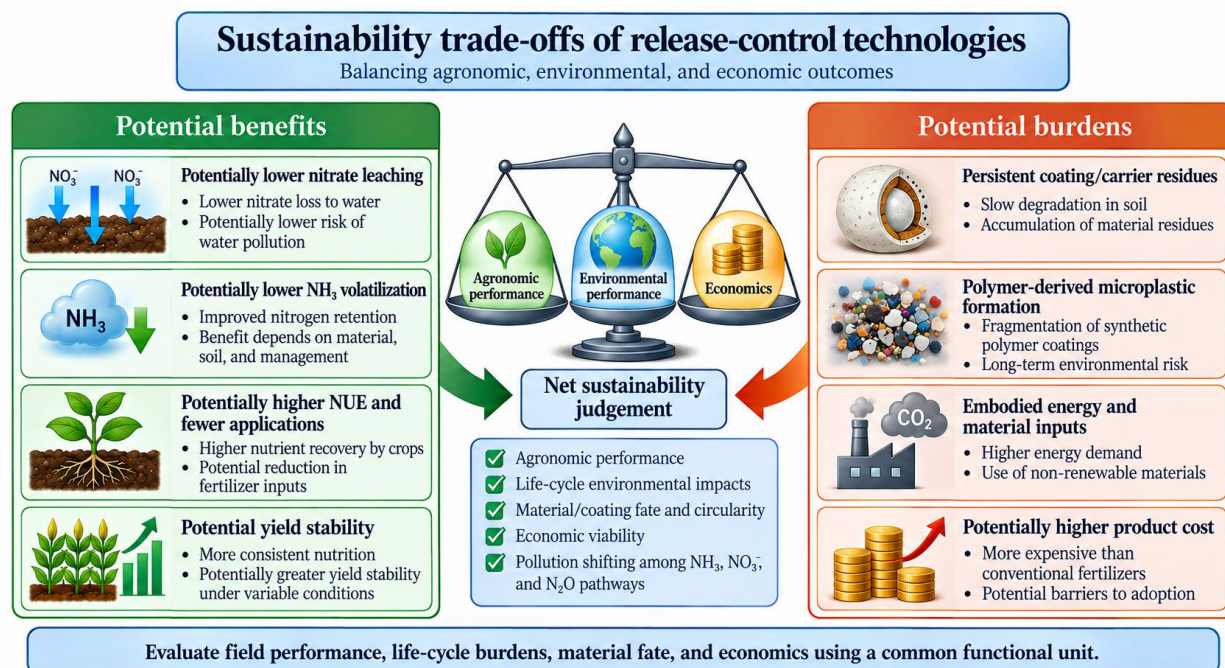


Figure 7. Sustainability trade-offs of release-control fertilizer technologies.

Note. Trade-off framework adapted by the author from Bhattacharjee et al. (2025), Li et al. (2025), Tian et al. (2024), and Tian et al. (2026).

10.4. Life-cycle performance

Life-cycle assessment (LCA) is therefore necessary when comparing material-intensive CRFs with conventional fertilizers. LCA-based work on lignin-derived systems and performance-first carbonaceous fertilizers illustrates the need to integrate raw-material origin, synthesis energy, coating mass, field efficiency, avoided nutrient losses, and end-of-life behavior (Ji et al., 2024; Pereira, 2026). A fertilizer cannot be classified as environmentally superior solely because nutrient losses are reduced. Table 7 summarizes the principal benefits and potential burdens that should be assessed together. Accordingly, LCA should be interpreted as a multi-endpoint, functional-unit

comparison rather than a single indicator: climate change, eutrophication/acidification, resource and energy demand, coating persistence or fragmentation, field nutrient-loss pathways, and avoided application burdens should be evaluated together when data permit.

Table 7. *Environmental benefits and potential environmental burdens associated with release-control technologies.*

Dimension	Potential benefit	Potential burden / uncertainty
N leaching	Lower early mineral-N surplus	Late-season release can leave residual N.
NH ₃ volatilization	Reduced urea exposure at high concentration	Benefit can shift N toward other pathways.
N ₂ O	Potential reduction with better synchrony	Can increase under some moisture/N conditions.
P loss	Lower soluble-P peaks with sparingly soluble sources	Insufficient dissolution can limit crop P.
Coating fate	Biobased/photodegradable options are emerging	Fragmentation and microplastic residues remain possible.
Carbon footprint	Higher NUE can reduce fertilizer requirement	Coating synthesis adds material and energy burdens.
Operations/economics	Fewer applications can lower labor and traffic	Higher product price and manufacturing complexity.

Note. Net environmental benefit requires a functional-unit comparison that includes fertilizer manufacture, field performance, nutrient losses, and coating fate.

11. Manufacturing, Scale-Up, Economics, and Industrial Readiness

11.1. Coating and matrix manufacturing

Commercial viability requires reproducible manufacturing at throughput, not only reproducible chemistry at bench scale. Coating technologies include pan coating, fluidized-bed and Wurster processes, spray application, hot-melt encapsulation, extrusion, and matrix formation. Process variables such as atomization, solids content, particle circulation, drying, coating mass,

viscosity, and residence time affect coating uniformity and therefore release variability (Ali et al., 2023; Azeem, 2025; Azeem, 2026; Bi et al., 2020; Nguyen et al., 2021; Shi et al., 2026; Su et al., 2024; Uzoh et al., 2026).

11.2. Mechanical integrity and handling

Scale-up creates additional constraints often absent from materials studies. Granule attrition and breakage can expose soluble cores; coating defects can increase as particle-size distributions broaden; heat and mass transfer can change with bed hydrodynamics; and storage or application equipment can damage mechanically weak products. Recent work on fertilizer-distributor breakage shows that product design must account for downstream handling (Zhang, Chen, et al., 2026).

11.3. Cost, raw materials, and industrial constraints

Industrial readiness also depends on cost and supply chains. Low-cost wax coatings, waste-derived copolymers, fly-ash geopolymers, pilot-scale zeolites, and recovered-nutrient routes illustrate attempts to couple release control with inexpensive or circular feedstocks (Bonetti et al., 2022; Ghumman et al., 2023; Hamidi et al., 2022; Hu et al., 2024; Shi et al., 2026; Su et al., 2024). Patent landscapes show that commercially relevant controlled-release urea technologies are strongly shaped by formulation simplicity, scalable coating routes, and product specifications, not only by laboratory novelty (Liu et al., 2022). Table 8 compares the principal manufacturing routes by scale-up strengths, process risks, and qualitative industrial-readiness signals. In this review, "commercial promise" denotes a technically plausible path toward deployment, whereas "demonstrated commercial feasibility" is reserved for technologies with repeatable scaled

production, product-quality consistency, handling/storage stability, credible techno-economic performance, and an identifiable regulatory and market pathway.

Table 8. *Manufacturing routes, scalability, main process risks, and qualitative industrial-readiness signals.*

Route	Scale-up strengths	Primary scale-up risks	Industrial-readiness signal
Pan coating	Simple equipment; broad familiarity	Nonuniform residence and coating distribution	Suitable where tight uniformity is not essential.
Fluidized-bed/Wurster	Good heat/mass transfer and coating control	Hydrodynamics, nozzle scaling, attrition	Strong route for engineered CRU if reproducible.
Spray coating	Continuous potential	Overspray, drying load, droplet-size control	Depends on solvent/binder system.
Hot-melt encapsulation	No solvent recovery; compact process	Melt rheology and thermal compatibility	Promising for wax/thermoplastic coatings.
Extrusion/matrix formation	Continuous and scalable	Thermal/mechanical damage and nutrient distribution	Good for matrix-type SRF composites.
Mineral activation/granulation	Uses low-cost mineral feedstocks	Energy intensity and variable reactivity	Readiness depends on beneficiation/activation economics.
Recovered-nutrient crystallization	Circular nutrient source	Feed purity and product consistency	Strong where wastewater infrastructure supports recovery.

Note. Qualitative readiness reflects process characteristics, not a certification of commercial maturity.

11.4. Technology-readiness framework (TRL-SRF)

Figure 8 proposes a five-level technology-readiness framework for slow- and controlled-release fertilizers (TRL-SRF): TRL-SRF 1, laboratory proof of concept; TRL-SRF 2, controlled-soil validation; TRL-SRF 3, pot/greenhouse agronomic validation; TRL-SRF 4, multi-location or multi-season field validation; and TRL-SRF 5, industrial/commercial readiness. TRL-SRF is a field-specific conceptual simplification developed for this review and is not equivalent to

conventional nine-level technology-readiness systems (Mankins, 1995). It should therefore not be interpreted as a certification standard or as a direct mapping onto conventional TRL numbers. Proposed gate criteria are as follows. Level 1 requires reproducible nutrient loading, a reproducible release profile, and basic physicochemical characterization. Level 2 requires soil incubation, testing under at least two relevant moisture/temperature or chemical conditions, and evidence supporting the proposed release mechanism. Level 3 requires replicated crop response, nutrient uptake/NUE measurements, and comparison with an appropriate conventional fertilizer control. Level 4 requires replicated field trials across at least two seasons or contrasting environments, reproducible agronomic performance, and measurement of at least one relevant nutrient-loss or environmental endpoint. Level 5 requires repeatable pilot/industrial production, product consistency, storage and handling integrity, preliminary techno-economic feasibility, and evidence of regulatory/commercial pathway readiness. These gates are proposed for reuse and independent validation; they are intentionally conservative and distinguish laboratory novelty from manufacturing and market readiness.

Figure 8 summarizes the TRL-SRF scale and the validation gates needed to progress from material concept to commercial fertilizer.

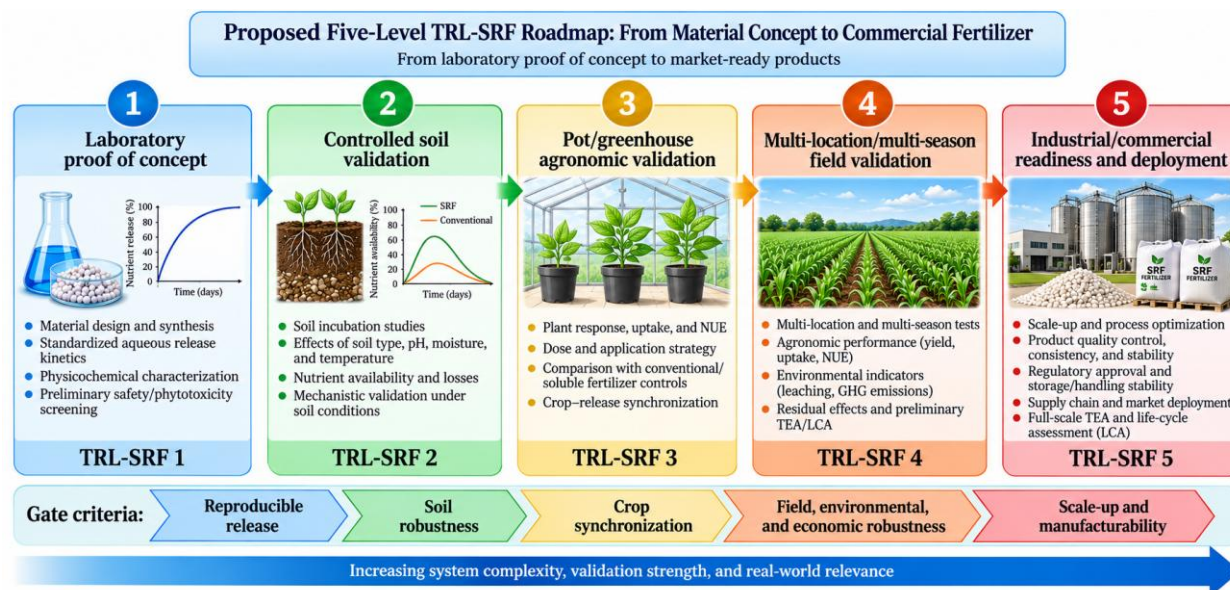


Figure 8. Roadmap from material development to commercially viable fertilizer using the proposed TRL-SRF scale.

Note. TRL-SRF roadmap proposed by the author and informed by scale-up, patent, and coating-process evidence from Liu et al. (2022), Nguyen et al. (2021), and Uzoh et al. (2026).

12. Critical Gaps and Research Priorities

12.1. Release and mechanism gaps

The first research gap is a release-characterization gap. A large fraction of formulations are still evaluated primarily in water, despite evidence that placement, temperature, soil texture, ionic composition, and mineral reactions can materially alter release (El Messbahi et al., 2026; Li et al., 2024; Ransom et al., 2020; Roy et al., 2023; Swify et al., 2023; Teles et al., 2020). Standardized protocols should therefore include both a reference aqueous test and at least one soil-relevant test matrix.

The second gap is mechanistic. Empirical equations remain useful descriptors, but mechanistic interpretation should be supported by changes in coating morphology, matrix

structure, speciation, or mineral phase. Combining release experiments with modeling, synchrotron characterization, molecular dynamics, and structural analysis provides a stronger basis for mechanism attribution (El Messbahi et al., 2026; Lakshani et al., 2023; Lipin et al., 2023; Mendonca Cidreira et al., 2026; Mihok et al., 2020).

12.2. Scale and duration gaps

The third gap is scale and duration. Pilot manufacturing, multi-year field trials, and industrial validation remain much rarer than material synthesis studies. The most decision-relevant evidence includes long-term nutrient cycling, soil-quality response, multi-season crop response, product breakage, and manufacturing reproducibility (Bonetti et al., 2022; Gao et al., 2024; Li et al., 2020; Liu, Chen, et al., 2025; Nguyen et al., 2021; Tian et al., 2026; Wang et al., 2024; Zhang, Chen, et al., 2026).

12.3. Economic and sustainability gaps

The fourth gap is economic. Cost is commonly discussed qualitatively, but few studies report cost per kilogram of nutrient effectively absorbed, cost per unit of yield gain, or the value of avoided nutrient loss. Future techno-economic analysis should integrate coating yield, energy, raw materials, application frequency, product damage, field efficacy, and residual soil value. The same principle applies to circular materials: waste-derived feedstocks are not automatically low-cost if purification, activation, or process complexity is high (Ghumman et al., 2023; Hamidi et al., 2022; Hu et al., 2024; Shi et al., 2026; Su et al., 2024). Techno-economic analysis (TEA) should report transparent functional metrics such as cost per kilogram of nutrient delivered, cost per kilogram of nutrient effectively absorbed, cost per unit of yield gain, coating/material cost per tonne of product, energy demand, manufacturing yield, and sensitivity to throughput and raw-material price. When

possible, avoided application events, avoided nutrient losses, residual soil value, and end-of-life costs should be included so that a low laboratory material cost is not confused with a competitive delivered fertilizer cost.

The fifth gap is the lack of standardization in sustainability claims. Biobased coatings, biodegradable polymers, hydrogels, mineral matrices, and recovered-nutrient fertilizers require distinct end-of-life metrics. Persistence, fragmentation, ecotoxicity, greenhouse-gas effects, and nutrient-loss reduction should be reported in compatible functional units so that environmental benefits are not shifted from one impact category to another (Bhattacharjee et al., 2025; Ji et al., 2024; Jo et al., 2026; Li et al., 2025; Pereira, 2026; Tian et al., 2026; Tian et al., 2024).

12.4. Research priorities

Finally, next-generation research should shift from maximizing release duration to optimizing nutrient-delivery synchronization. That change would align formulation science with crop physiology, soil process dynamics, and industrial decision-making. Table 9 summarizes the priority gaps, the evidence level required to close them, and recommended research actions.

Table 9. Research gaps, evidence required to close them, and priorities for future slow- and controlled-release fertilizer research.

Gap	Current weakness	Evidence needed	Priority
Release characterization	Water tests dominate	Soil-relevant standardized release protocols	Very high
Mechanism	Curve fit used as proof	Structural/speciation evidence + validated models	Very high
Synchronization	Release rarely matched to crop demand	Field-calibrated R(t) and D(t) curves	Very high
Soil variability	Single-soil conclusions generalized	Multi-soil and climate validation	High

Gap	Current weakness	Evidence needed	Priority
Duration	Few multi-season studies	Residual fertility and long-term environmental data	High
Scale-up	Bench synthesis overrepresented	Pilot throughput, uniformity, attrition, storage data	Very high
Economics	Cost discussed qualitatively	Cost per kg of nutrient absorbed and techno-economic analysis (TEA)	High
Sustainability	Single impact endpoints	LCA including coating fate and microplastics	Very high

Note. Priority reflects the extent to which the gap limits mechanistic interpretation, field transferability, environmental assessment, or commercialization.

12.5. Limitations of the Review

This review has several boundary conditions that should temper interpretation. First, it is a curated critical integrative review rather than a prospectively registered systematic review, and raw initial-hit, screening, and exclusion counts were not logged in real time. Second, Google Scholar provides broad interdisciplinary coverage but has variable indexing transparency, and the final corpus is predominantly English-language. Third, publication bias may favor successful formulations, while negative pilot-scale or industrial results are often proprietary or unpublished. Fourth, publication growth during 2025–2026 means that the evidence base is evolving rapidly. Fifth, terminology and test protocols remain inconsistent across studies, particularly for release duration, biodegradability, NUE, and environmental endpoints. Sixth, the evidence base is nutrient-imbalanced: N is strongly represented, P is moderate, K/mineral systems are more heterogeneous, and micronutrient evidence is limited. Seventh, geographic concentration and climate/soil specificity constrain direct transfer of some field results. Finally, the review was screened and synthesized by a single author; Supplementary File S1 and the explicit evidence hierarchy are provided to improve auditability, but independent replication of the selection and coding process would further strengthen confidence.

13. Conclusions and Outlook

13.1. Conditions for real advantage

Slow- and controlled-release fertilizers are most valuable when release control resolves a real temporal mismatch between nutrient supply and crop demand without introducing an offsetting environmental or economic burden. The evidence does not support a universal ranking that makes the slowest formulation best. Performance depends on nutrient chemistry, release mechanism, soil and climate, crop phenology, placement, management, product integrity, and the feasibility of manufacturing the formulation consistently.

13.2. Nutrient-specific implications

For nitrogen, the evidence base is strongest, and the clearest case arises where early losses are large and sustained supply can replace or simplify split applications. For phosphorus, the evidence is moderately developed and value depends on maintaining plant availability while reducing fixation or loss. For potassium, controlled soluble salts and slowly reactive mineral sources require fundamentally different evaluation criteria, and evidence remains more heterogeneous. Broad conclusions for micronutrients are not yet justified. Across nutrients, multi-season field evidence is more informative than aqueous release tests because it captures crop uptake, residual fertility, soil interactions, weather variability, and management effects.

13.3. Outlook

The central conclusion of this review is therefore performance-first: the optimum fertilizer is not the formulation with the slowest nutrient release, but the formulation whose nutrient-delivery profile most closely matches crop demand under realistic soil and climatic conditions while minimizing environmental and economic burdens. Future progress will depend on integrating

mechanistic release science with field-calibrated synchronization, standardized sustainability metrics, techno-economic analysis, and manufacturing validation.

Declarations

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Conflict of Interest

The author declares no conflict of interest.

Data Availability

No new experimental datasets were generated. All quantitative values discussed in the review are traceable to the cited literature. Supplementary File S1 provides the archived Boolean-search audit, the deduplicated bibliographic matrix, and the reference-level evidence register used to improve review auditability.

Author Contributions

Conceptualization, critical analysis, literature synthesis, visualization, and manuscript preparation by the author.

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